ENCODE LONDON 2026 · SOLO BUILD

SUBMISSION · 4 TRACKS

TUBESENSE

Trains located between stations from ordinary phone rides, and a cited evidence pack. No GPS. Every claim checked. A human signs.

Tobi Ojebiyi · github.com/coderback

87 dB

TfL’s own in-cab LAeq, Liverpool Street → Bethnal Green, Central line eastbound.

Measured 24 June 2024, released under FOI-1231-2425.

83 dB

Bethnal Green → Mile End, same TfL measurement

85 dB(A)

HSE upper exposure action value, for reference

92.8 dB

Reported in the press, Jan 2026. Primary source not located

Sources: TfL FOI-1231-2425 · Brixton Buzz, 27 Jan 2026 (secondary)

TFL: CORRUGATION COMES BACK AFTER GRINDING

Dec 2021 – May 2022

Both sections were ground for noise (direction not stated).

Mayor’s Question 2022/1791

Every 6–12 months

TfL says grinding must be repeated, “usually 6 months – a year”.

TfL Tube Noise presentation, 6 Dec 2022

16 Jul 2026

TfL turned down most of Elly Baker AM’s data requests.

Mayor’s Question Time 2026/2156

Schematic, not to scale: corrugation = periodic ripples worn into the rail head

03 · WHO IT’S FOR

ELLY BAKER AM

Labour transport spokesperson · Deputy Chair, London Assembly Transport Committee

> At Mayor’s Question Time on 16 July 2026 she said TfL had turned down most of her requests for noise data, and plans grinding only eight weeks ahead.

WHAT SHE NEEDS

1. Phone measurements placed between stations, without GPS
2. A cited pack that says what the record does and does not show
3. The precise information requests to make next

Ultimately for those exposed longest: train drivers and their unions (RMT, ASLEF) · MQT 2026/2156

ONE RIDE IN, A SIGNED DRAFT OUT

01
RIDE
A phone on a seat: accelerometer, gyro, magnetometer, barometer, audio.

02
CLEAN
Sync, pauses, a phone on the floor, split runs: flagged from the data.

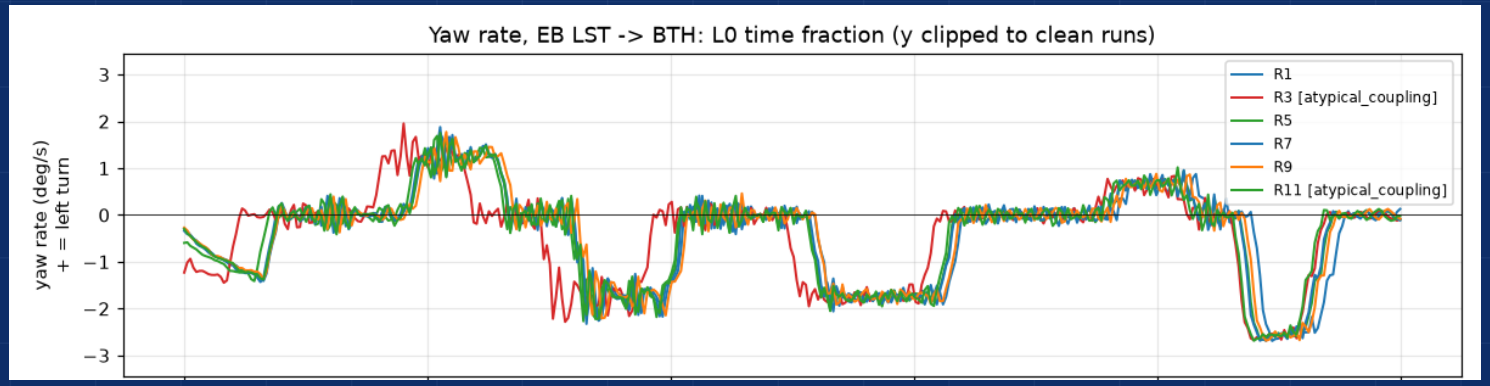
03
LOCATE
Curve signatures place the train between stations. No GPS.

04
MEASURE
1-second A-weighted levels and bands. Raw audio never leaves the laptop.

05
INVESTIGATE
Gemini 4 Argon reads TfL and Assembly records from MongoDB Atlas.

06
APPROVE
Code checks every quote and number. A human signs the draft.

EVERY CURVE IS A FINGERPRINT



REAL DATA, 6 RIDES

Gyro yaw rate along Liverpool St → Bethnal Green. The same curves arrive in the same order on every ride: that repeat is the fingerprint the tracker matches.

r 0.76–0.99

Gyro yaw traces of the same section, ride to ride (Central, 23 Oct). Different sections: 0.20.

DTW

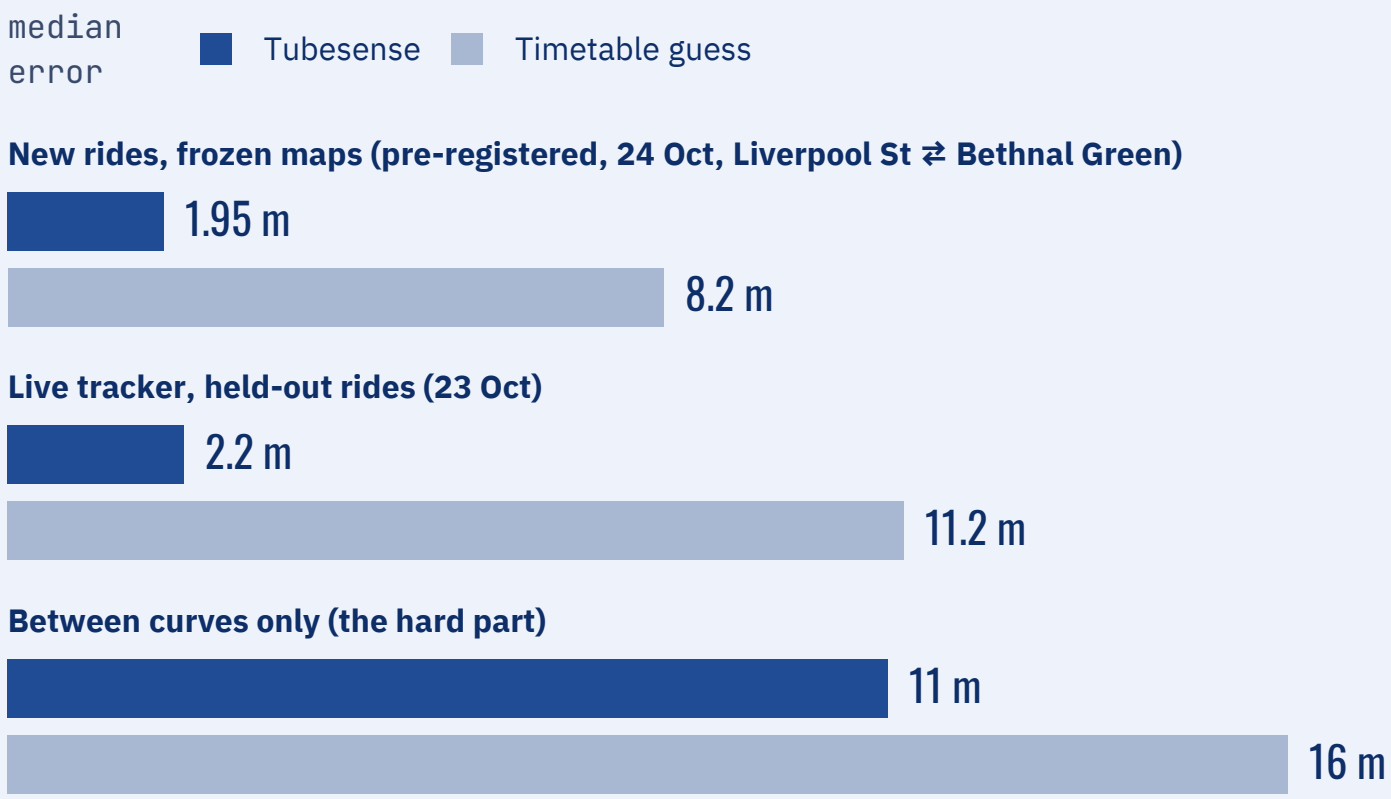
Matches a new ride to a map of earlier rides in distance, not time, between two stops.

Gyro wins

Ablation: the magnetometer adds nothing to positioning (NO-GO as a fingerprint).

Real data: outputs/gonogo/gyro_EB_LST-BTH.png · outputs/FINDINGS.md (a), (b), (e)

METRES, NOT STATIONS



96%

of the time the truth sat inside the tracker's 90% band on new rides.

Error = agreement with curve-edge landmarks matched across rides; there is no GPS truth in a tunnel

WE PUBLISH THE NO-GOS TOO

Question	Verdict	Evidence
Do gyro curve signatures repeat?	GO	r 0.76–0.99 same section vs 0.20 different
Positioning on new rides (pre-registered)	GO	1.95 m vs 8.2 m timetable, coverage 0.96
Does it transfer to Hammersmith & City?	NO-GO	One usable pair; 52 m vs 47 m timetable
Rank section loudness from phone audio	NO-GO	10 of 11 trips, +0.54 dB mean; rule needed all 11
Magnetometer as a position fingerprint	NO-GO	No stable platform signature (sd 19 vs 22 µT)
Absolute level, NIOSH app (indicative)	INDICATIVE	91.9 dB(A) EB, 87.3 WB, whole trip; app valid to ~85
Corrugation wavelength from the roar	NO-GO	Peaks don’t move with speed (9/9); cause untested

THE AGENT CAN'T MARK ITS OWN HOMEWORK

PROPOSE

Argon picks tracker settings from an allow-list, each with a stated hypothesis.

SCORE

A harness it cannot edit scores each one, with a rule fixed before the session.

LOG

Every call is hash-chained and mirrored to MongoDB Atlas.

ONE SHOT

A single sealed score on rides recorded after it chose.

6 → 1

experiments → accepted

Sealed test: **1.8 m, the same as the old tracker.** The tuning gain didn't carry over, and the log proves we only got one try.

EVERY ARGON JOB HAS A CHECK

ASK

Live answers go through the verifier, then an Argon review; flagged claims are withheld.

READ

Reads TfL's 2018 roughness charts from the PDF; a pixel reader checks every value.

RE-DERIVE

Recomputes our headline numbers with its own code: 65 of 65 match.

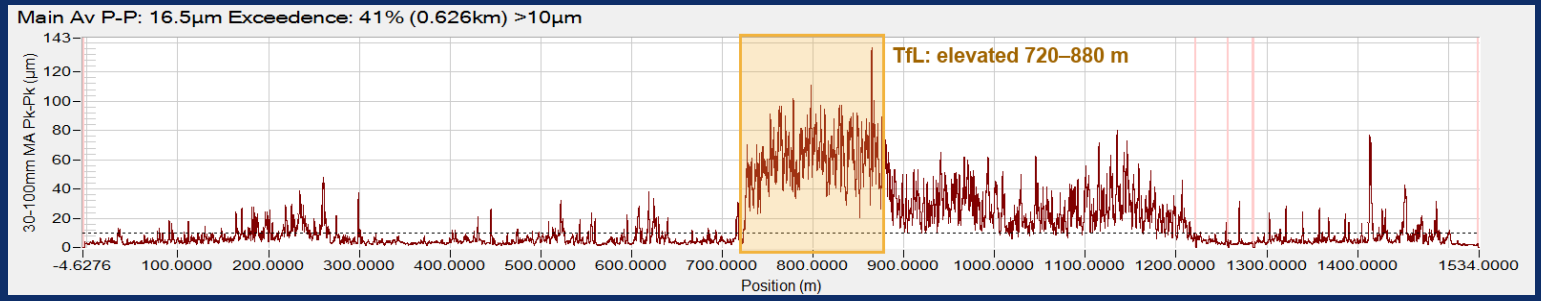
COMPARE

Flash vs Argon, 6 runs each: equal on the verifier; Argon cites more of the record.

120/120

chart readings match the pixels

Checks still caught misnamed colours and invented numbers. Pending a human check.



TfL FOI-1730-1819, Jan 2018, Bethnal Green → Mile End EB. Highlight: TfL's own stated stretch.

10 · INVESTIGATOR + MONGODB ATLAS

TFL'S OWN RECORDS, SEARCHED THREE WAYS

14 PUBLIC DOCUMENTS · 44 CHUNKS

Atlas Search (full-text)	LIVE
Automated Embedding · Voyage 4	LIVE
Vector Search	LIVE
Hybrid Search ($rankFusion)	LIVE
Stream Processing	LIVE

12 / 12

test questions where hybrid search ranks the expected document first (our questions, not blind)

Verifier canary (seeded claim)

"92.8 dB is TfL's own data" → REJECTED

ONE CLUSTER, TWO PIPELINES

amber outline = runs in Atlas

LIVE RIDE DATA

A ride
Replayed through the tracker now, or live from the phone (phyphox)

→

ingest_live
One row per second, random trip id, no audio

→

Stream Processing
10 s tumbling windows, SP2 processor

→

live_view
Read by the web app as windows close

INVESTIGATOR RETRIEVAL

Public records
14 TfL / GLA documents, 44 chunks

→

Search + Vector
BM25 full text plus autoEmbed vectors (Voyage 4)

→

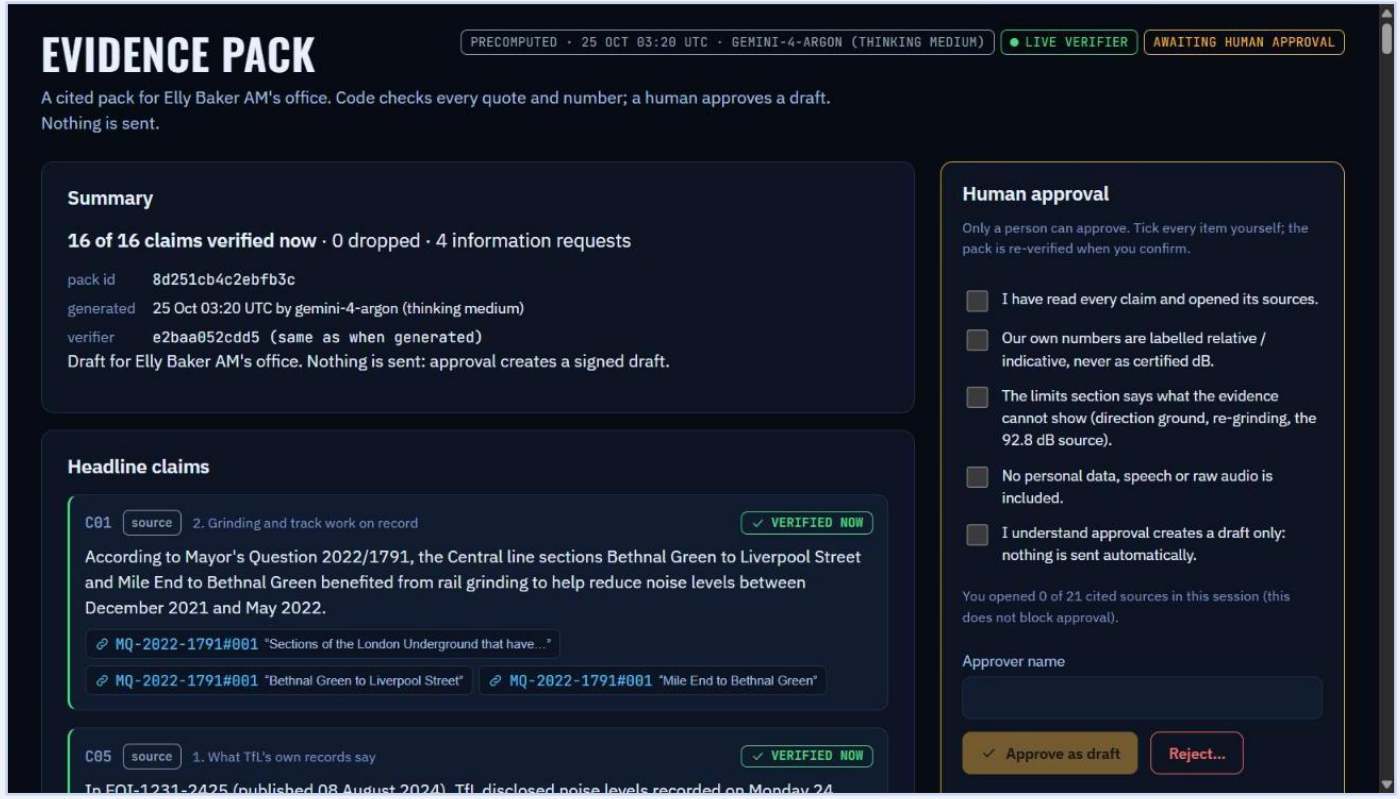
Hybrid
$rankFusion of both rankings

→

Gemini 4 Argon
Cited claims, then the deterministic verifier

Also in Atlas: the hash-chained attempt logs, claims, approvals, asks, roughness, rechecks · derived data and public text only

NOTHING IS SENT AUTOMATICALLY



RE-VERIFY

The verifier runs again at the moment of approval.

SIGN

SHA-256 over pack, approver, time and verifier version.

RECORD

Written to Atlas *approvals*; rejections are kept with a reason.

Web app Evidence pack page · agent/approve.py · pack 8d251cb4c2ebfb3c, 16 verified claims

REAL RIDES ARE MESSY. PRIVACY IS NOT OPTIONAL.

DETECTED FROM THE DATA, NOT HAND-CODED

- A phone on the floor, from its tilt and vibration
- Two app pauses, recovered from phyphox's event log
- A trip split across two files after a wrong stop
- Calibrated vs raw magnetometer (it reversed our own notes)
- Silent sync taps, so it falls back to matching vibration to sound

PRIVACY BY DESIGN

- Audio is decoded in memory and never written to disk
- Only 1-second levels and third-octave bands are kept
- Gemini sees public documents and summary statistics only
- Raw recordings deleted on 1 Nov 2026

> The ICO calls recording conversations "highly intrusive", so we keep no audio at all.

2 GNSS ANTENNA
Active patch · SMA

1 TOP SHELL
Vented lid · 4 × M3

13 POWER PACK
USB-C PD · strapped

4 RASPBERRY PI 5
Edge compute · active cooler

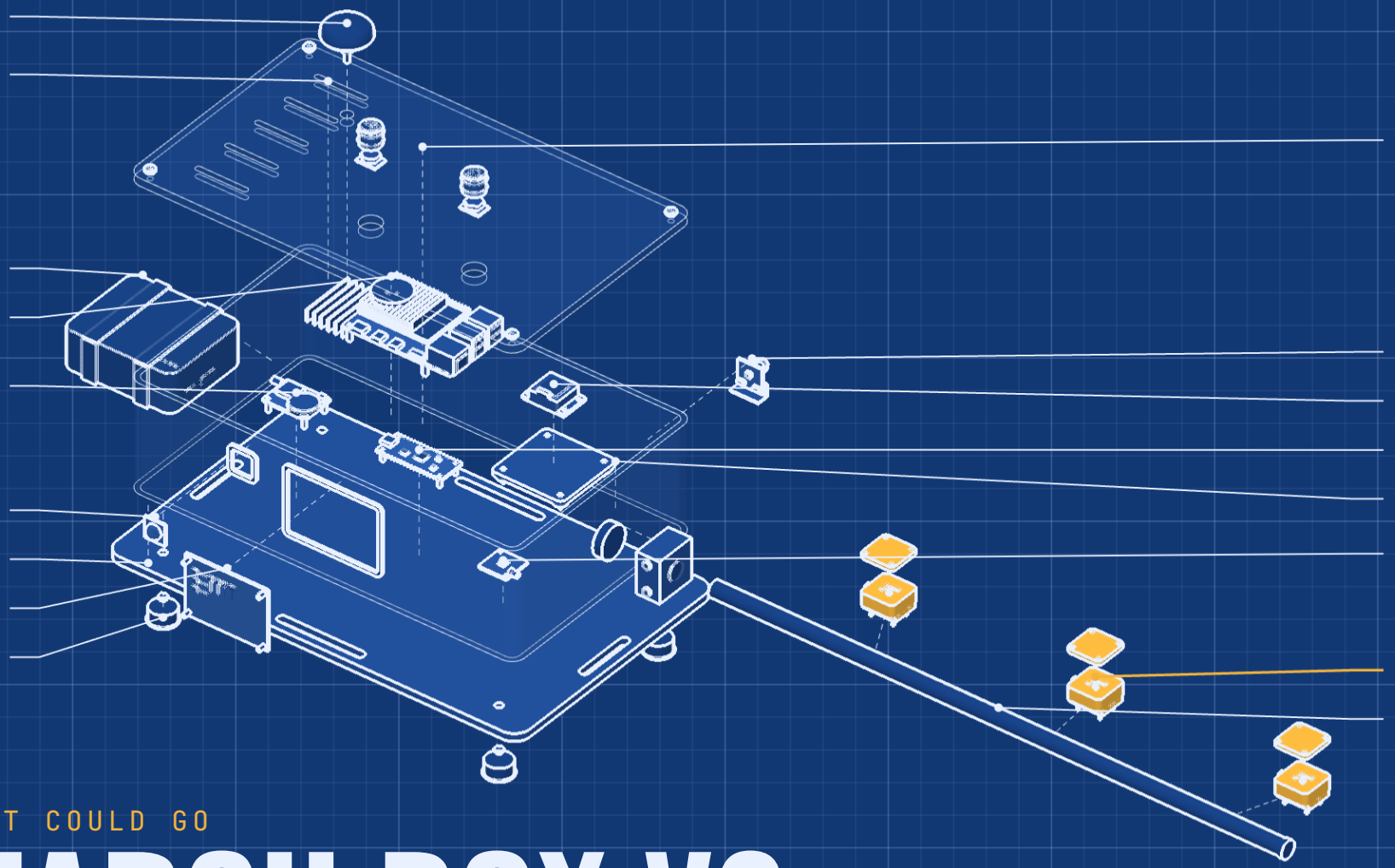
6 GNSS RECEIVER + RTC
u-blox · DS3231 · CR2032

12 LIGHT SENSOR
VEML7700

16 BASE PLATE
6 mm aluminium

11 MMWAVE RADAR
TI IWR6843 · 60 GHz

17 ISOLATOR FEET
4 × rubber mounts



MEMS MICROPHONES ×2 3
TDK ICS-43434 · I²S

FAST BAROMETER 10
BMP585 class · vented

INDUSTRIAL IMU 7
ADIS16470 class

SENSOR HUB 5
RP2040 · hardware timestamps

ISOLATED SUB-PLATE 8
4 × silicone grommets

VIBRATION ACCELEROMETER 9
ADXL1002 · stud-mounted

MAGNETOMETER PODS ×3 14
PNI RM3100 · 140 mm pitch

CARBON BOOM 15
Ø12 mm · 440 mm

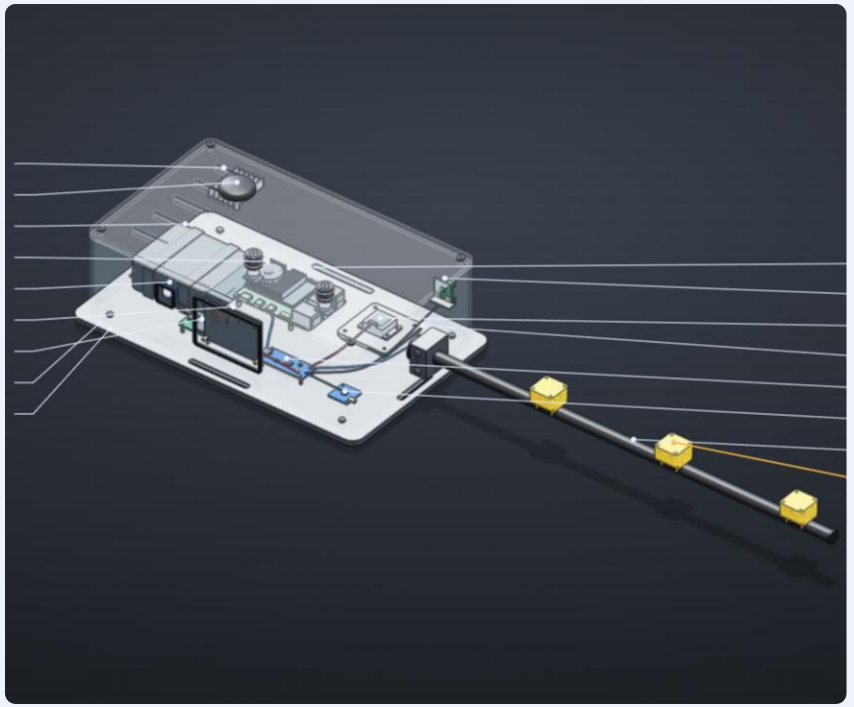
14 · WHERE IT COULD GO

RESEARCH BOX V2

CONCEPT · NOT BUILT

EACH PART TARGETS A LIMIT WE MEASURED

Signal	Phone today	Research Box v2
Timebase	Two apps, synced by a knock	GNSS PPS + RTC, one hardware clock
Acoustic	Voice Memos limiter, relative only	2 × I²S MEMS mics, calibratable dB(A)
Magnetic	One sensor beside the battery	3 × RM3100 array on a boom (to test)
Speed	Inferred from position	Radar Doppler (to test) + sleeper frequency
Inertial	Phone IMU, ~99 Hz	ADIS16470-class IMU, isolated
Pressure	Phone barometer	BMP585-class, tens of Hz, vented



BUILT IN A WEEKEND, HONEST ABOUT THE REST

WORKING

- GPS-free tracker + web app
- Messy-input QC
- All 6 Atlas features, incl. streams
- Analyst harness + hash-chained log
- Investigator, Ask Argon, verifier, approval

IN PROGRESS

- Signed evidence pack (live in demo)
- Live phone sensors via phyphox
- Human check of roughness readings
- 3-minute demo video

NEXT

- More rides, more lines
- A calibrated reference meter
- Our noise vs TfL roughness plots
- Station labels via Gemini Transcribe
- Research Box v2 prototype

ORDINARY RIDES.
NO GPS.
TFL'S OWN RECORDS.
EVERY CLAIM CHECKED.
A HUMAN SIGNS.

Raw audio never leaves the laptop.